

## SAPPORO, Japan

WMO: 474120

Lat: 43.0600N

Lon: 141.3286E

Elev: 26

StdP: 101.01

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |       |     |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-------|-----|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%   |     |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP    | HR  | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)   | (h) | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | -9.9       | -8.6 | -17.0                         | 0.8 | -7.8 | -15.3 | 1.0 | -6.8 | 11.6                  | 0.8  | 9.7 | -0.5 | 2.3                   | 170  | 0.514 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 6.5                    | 29.5            | 22.8 | 27.9 | 21.8 | 26.5 | 20.9 | 23.8                | 28.1 | 23.0 | 26.6 | 22.1 | 25.4 | 3.8                  | 150  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 22.4                            | 17.2 | 26.4 | 21.6 | 16.3 | 25.6 | 20.8 | 15.5 | 24.8 | 71.7          | 28.2 | 68.4 | 26.7 | 65.2 | 25.5 | 27.2              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| (4) 10.1          | 8.7  | 7.5 | -12.1                      | 32.5 | 1.7                | 1.8 | -13.3                                              | 33.8 | -14.2      | 34.8 | -15.2      | 35.8 | -16.4      | 37.1 |    |    |
| (5)               |      |     | -12.9                      | 24.8 | 1.6                | 1.2 | -14.0                                              | 25.7 | -15.0      | 26.3 | -15.8      | 27.0 | -17.0      | 27.8 |    |    |

## Monthly Climatic Design Conditions

|      |                                                                                  | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |
|------|----------------------------------------------------------------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|      |                                                                                  | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |
| (6)  | Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 9.4   | -3.5  | -2.9  | 1.0   | 7.4   | 13.3  | 17.4  | 21.6  | 22.8  | 18.8  | 12.2  | 5.1   | -1.0 | (6)  |
| (7)  |                                                                                  | DBStd       | 9.75  | 2.71  | 3.05  | 3.22  | 3.62  | 3.34  | 2.98  | 2.67  | 2.59  | 2.96  | 3.18  | 3.99  | 3.10 | (7)  |
| (8)  |                                                                                  | HDD10.0     | 1674  | 419   | 362   | 279   | 95    | 9     | 0     | 0     | 0     | 0     | 15    | 153   | 342  | (8)  |
| (9)  |                                                                                  | HDD18.3     | 3569  | 677   | 595   | 538   | 327   | 159   | 52    | 4     | 1     | 28    | 191   | 397   | 600  | (9)  |
| (10) |                                                                                  | CDD10.0     | 1460  | 0     | 0     | 0     | 17    | 112   | 222   | 359   | 396   | 265   | 83    | 6     | 0    | (10) |
| (11) |                                                                                  | CDD18.3     | 314   | 0     | 0     | 0     | 0     | 3     | 24    | 105   | 138   | 43    | 1     | 0     | 0    | (11) |
| (12) |                                                                                  | CDH23.3     | 1468  | 0     | 0     | 0     | 1     | 31    | 101   | 463   | 714   | 157   | 1     | 0     | 0    | (12) |
| (13) |                                                                                  | CDH26.7     | 292   | 0     | 0     | 0     | 0     | 4     | 13    | 100   | 156   | 19    | 0     | 0     | 0    | (13) |
| (14) | Wind                                                                             | WSAvg       | 3.3   | 3.0   | 3.2   | 3.5   | 3.8   | 3.8   | 3.4   | 3.2   | 3.1   | 3.0   | 3.2   | 3.1   | 2.9  | (14) |
| (15) | Precipitation                                                                    | PrecAvg     | 1145  | 109   | 93    | 80    | 61    | 56    | 64    | 81    | 135   | 136   | 114   | 106   | 109  | (15) |
| (16) |                                                                                  | PrecMax     | 1674  | 182   | 268   | 157   | 133   | 116   | 169   | 194   | 644   | 314   | 256   | 266   | 209  | (16) |
| (17) |                                                                                  | PrecMin     | 728   | 50    | 28    | 32    | 6     | 14    | 6     | 16    | 37    | 31    | 19    | 43    | 53   | (17) |
| (18) |                                                                                  | PrecStd     | 169   | 32    | 43    | 31    | 32    | 23    | 36    | 43    | 98    | 70    | 49    | 44    | 36   | (18) |
| (19) | Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB    | 4.6   | 6.8   | 12.2  | 22.0  | 26.8  | 28.4  | 31.4  | 32.4  | 28.9  | 22.2  | 16.8  | 10.2 | (19) |
| (20) |                                                                                  |             | MCWB  | 2.5   | 3.6   | 6.6   | 11.6  | 15.8  | 19.3  | 23.9  | 24.3  | 22.4  | 15.7  | 11.7  | 7.8  | (20) |
| (21) |                                                                                  | 2%          | DB    | 3.0   | 4.7   | 9.5   | 18.1  | 23.5  | 25.9  | 29.2  | 30.0  | 26.7  | 20.1  | 14.9  | 7.4  | (21) |
| (22) |                                                                                  |             | MCWB  | 1.1   | 1.8   | 4.8   | 10.6  | 14.6  | 18.3  | 22.6  | 23.6  | 20.4  | 14.8  | 10.7  | 4.9  | (22) |
| (23) |                                                                                  | 5%          | DB    | 1.8   | 3.2   | 7.5   | 15.3  | 21.0  | 24.2  | 27.5  | 28.3  | 25.1  | 18.7  | 13.0  | 5.4  | (23) |
| (24) |                                                                                  |             | MCWB  | -0.1  | 0.9   | 3.9   | 8.9   | 13.2  | 17.6  | 21.9  | 22.4  | 19.7  | 13.9  | 9.5   | 3.0  | (24) |
| (25) |                                                                                  | 10%         | DB    | 0.7   | 2.0   | 5.9   | 13.0  | 18.8  | 22.4  | 25.9  | 26.9  | 23.7  | 17.3  | 11.3  | 3.6  | (25) |
| (26) |                                                                                  |             | MCWB  | -1.1  | -0.2  | 2.8   | 7.8   | 12.2  | 16.8  | 20.9  | 21.6  | 18.9  | 13.1  | 8.2   | 1.3  | (26) |
| (27) | Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB    | 3.0   | 4.2   | 7.2   | 13.2  | 17.1  | 20.7  | 24.7  | 25.2  | 23.4  | 17.6  | 13.1  | 8.1  | (27) |
| (28) |                                                                                  |             | MCDWB | 4.5   | 6.2   | 10.9  | 19.3  | 25.0  | 26.1  | 29.6  | 30.3  | 27.4  | 19.9  | 15.3  | 9.8  | (28) |
| (29) |                                                                                  | 2%          | WB    | 1.4   | 2.3   | 5.5   | 11.2  | 15.4  | 19.3  | 23.5  | 24.2  | 22.2  | 16.0  | 11.4  | 5.3  | (29) |
| (30) |                                                                                  |             | MCDWB | 2.7   | 4.0   | 8.5   | 16.8  | 21.4  | 24.3  | 28.0  | 28.5  | 25.3  | 18.9  | 14.1  | 7.3  | (30) |
| (31) |                                                                                  | 5%          | WB    | 0.3   | 1.2   | 4.3   | 9.7   | 14.2  | 18.3  | 22.5  | 23.4  | 21.1  | 14.8  | 9.8   | 3.2  | (31) |
| (32) |                                                                                  |             | MCDWB | 1.6   | 3.0   | 7.0   | 14.3  | 19.2  | 22.8  | 26.4  | 27.1  | 23.9  | 17.8  | 12.6  | 5.0  | (32) |
| (33) |                                                                                  | 10%         | WB    | -1.0  | 0.0   | 3.1   | 8.4   | 13.3  | 17.4  | 21.6  | 22.6  | 19.7  | 13.6  | 8.4   | 1.5  | (33) |
| (34) |                                                                                  |             | MCDWB | 0.6   | 1.8   | 5.7   | 12.3  | 17.6  | 21.4  | 25.0  | 26.0  | 22.7  | 16.7  | 11.1  | 3.3  | (34) |
| (35) | Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR  | 5.3   | 5.8   | 6.1   | 7.5   | 8.1   | 7.5   | 6.8   | 6.5   | 7.3   | 7.5   | 6.3   | 5.2  | (35) |
| (36) |                                                                                  |             | MCDBR | 6.3   | 7.4   | 8.6   | 11.7  | 12.4  | 10.6  | 8.9   | 8.3   | 8.4   | 9.0   | 9.2   | 7.4  | (36) |
| (37) |                                                                                  |             | MCWBR | 5.5   | 6.0   | 5.9   | 6.4   | 6.1   | 5.1   | 4.2   | 3.8   | 4.6   | 5.5   | 6.7   | 6.2  | (37) |
| (38) |                                                                                  | 5% WB       | MCDBR | 6.4   | 7.2   | 8.1   | 10.5  | 10.3  | 9.0   | 7.8   | 7.2   | 6.8   | 7.8   | 8.7   | 7.4  | (38) |
| (39) |                                                                                  |             | MCWBR | 6.1   | 6.3   | 6.1   | 6.2   | 5.6   | 4.7   | 3.9   | 3.5   | 4.3   | 5.5   | 7.0   | 6.6  | (39) |
| (40) | Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.346 | 0.397 | 0.467 | 0.530 | 0.547 | 0.529 | 0.512 | 0.477 | 0.420 | 0.411 | 0.393 | 0.362 | (40) |      |
| (41) |                                                                                  | taud        | 1.984 | 1.840 | 1.746 | 1.764 | 1.823 | 1.947 | 2.062 | 2.177 | 2.287 | 2.167 | 2.113 | 2.001 | (41) |      |
| (42) |                                                                                  | Ebn at Noon | 736   | 756   | 752   | 743   | 746   | 761   | 770   | 785   | 805   | 748   | 685   | 671   | (42) |      |
| (43) |                                                                                  | Edh at Noon | 116   | 161   | 201   | 213   | 207   | 183   | 162   | 140   | 116   | 115   | 103   | 106   | (43) |      |
| (44) | All-Sky Solar<br>Radiation                                                       | RadAvg      | 1.28  | 2.03  | 3.26  | 4.51  | 5.15  | 5.02  | 4.56  | 4.36  | 3.70  | 2.62  | 1.48  | 1.04  | (44) |      |
| (45) |                                                                                  | RadStd      | 0.09  | 0.17  | 0.27  | 0.47  | 0.45  | 0.46  | 0.51  | 0.33  | 0.34  | 0.15  | 0.15  | 0.09  | (45) |      |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (46) | Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | +81     | N/A     |
| (47) | Regional (17 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page